MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL MX-9872(V)5/UA

This specification sheet is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the MX-9872(V)5/UA Scanner shall consist of this document and MIL-S-49166.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-9872(V)5/UA and augments MIL-S-49166. The MX-9872(V)5/UA is used on the M1 Tank Thermal Imaging System.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49166 to describe the MX-9872(V)5/UA Mechanical Scanner. All the MIL-S-49166 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49166 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49166 Scanner, Mechanical, MX-9872(V)/UA, General Specification for

USA TANK AUTOMOTIVE COMMAND

SC12271670 Product Fabrication Specification for M1 TIS Scanner, Mechanical Scan Interlace Assembly

DRAWINGS

USA ELECTRONICS R&D COMMAND DRAWINGS

80063 DLSM-B-773232, Scan-Interlace, PL-1408/UA

USA ARMAMENT R&D COMMAND DRAWINGS

19200 12271670, Scanner/Scan Position Sensor Assembly
19200 12272299, Mechanical Scanner Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Description. The MX-9872(V)5/UA Scanner, herein known as the MX-9872(V)5/UA, shall consist of the MIL-S-49166, MX-9872(V)/UA Scanner with scan position sensor assembly installed and adjusted for 41 Hz operation.

3.2 Construction. The MX-9872(V)5/UA shall be constructed in accordance with drawing 19200 12271670.

3.2.1 Weight. Weight of the MX-9872(V)5/UA shall be 1.90 pounds maximum.

3.5 Components.

a. Scanner, mechanical, MX-9872(V)/UA
b. Scan position sensor

3.6 Performance characteristics. The characteristics specified herein shall apply with the scan position sensor and associated hardware removed. Operation shall be at 41 Hz using the Scan-Interlace, PL-1408/UA (DLSM-B-773232).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Applicable for first article test only. See MIL-S-49166.

3.6.1.2 Specific scan frequency (41 Hz). Scan frequency of the MX-9872(V)5/UA shall be adjusted to 41.0, ±2.0 Hz. Scan frequency drift shall not be greater than ±2.1 Hz from the frequency at ambient setting (see MIL-S-49166, 4.4) when exposed to high and low temperature environmental conditions (see MIL-S-49166).

3.6.1.3 Synchronous scan frequency. Not applicable.

3.6.1.4 Scan angle. The scan angle shall be adjusted to 12.6, ±0.2 degrees.

3.6.2 Interlace. The MX-9872(V)5/UA shall operate in a 2:1 interlace mode. The interlace switching time shown in figure 1 of MIL-S-49166 shall not be greater than 3.6 milliseconds (ms).

3.6.3 Scan direction. The MX-9872(V)5/UA shall show the waveform shapes and phase relationships in figure 1 while operating in the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the MX-9872(V)5/UA shall not exceed 7.5 watts.

3.6.5 Scan efficiency. The MX-9872(V)5/UA scan efficiency shall be a minimum of 65 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 18.0 microseconds maximum.

3.7 Environmental conditions. See MIL-S-49166 except as follows:

3.7.5 Vibration. The MX-9872(V)5/UA shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 2 for the X axis and as shown in figure 3 of MIL-S-49166 for the Y and Z axes.

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During specified vibration, the average interlace waveform amplitude shown in MIL-S-49166, figure 1 shall not deviate from the amplitude without vibration by more than ±50 percent.

4. QUALITY ASSURANCE PROVISIONS

4.6 Test methods. Tests shall be performed utilizing the DLSM-B-773232 Scan-Interlace with the MX-9872(V)5/UA.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. See MIL-S-49166.

4.6.1.2 Specific scan frequency (41 Hz). The MX-9872(V)5/UA shall be connected to the Scan-Interlace DLSM-B-773232, and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace DLSM-B-773232) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (DLSM-B-773232) shall be adjusted to obtain the specified frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. Not applicable.

4.6.1.4 Scan angle. Scan angle measurement shall be in accordance with drawing 19200 12272299.

4.6.2 Interlace. Interlace measurement shall be in accordance with specification SC 12271670.

4.6.3 Scan direction. With the MX-9872(V)5/UA operating in either the 2:1 side or the 2:1 rear interlace mode, the outputs of the azimuth and interlace position transducers (Scan-Interlace DLSM-B-773232) shall be observed with an oscilloscope. The output waveforms shall show the applicable shape and phase relationships shown in figure 1. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49166.

4.6.5 Scan efficiency. See MIL-S-49166.

4.6.6 Scan jitter in azimuth. See MIL-S-49166.

4.7 Environmental tests. See MIL-S-49166 and figure 2.

6. NOTES

6.1 Intended use. The MX-9872(V)5/UA is intended for use with the Thermal Imaging System of the M1 Tank.

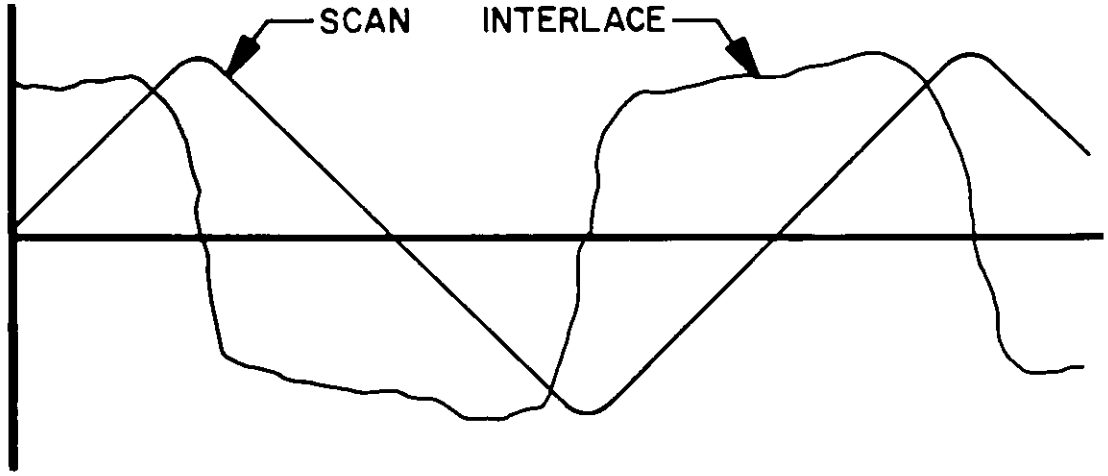
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2:1 REAR POSITION
INTERLACE LEADS SCAN APPROX. 90°



NOTES:

PL-1408/UA- CONNECT P1-33 TO P1-34 AND P1-4 TO P1-17

OSCILLOSCOPE- CONNECT CHANNEL 1 (SCAN POSITION) HIGH INPUT TO J3 (OR J2-7) AND LOW SCOPE TO GND

- CONNECT CHANNEL 2 (SCAN INTERLACE) HIGH INPUT TO J6 (OR J1-9) AND LOW SCOPE TO GND

FIGURE 1. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.

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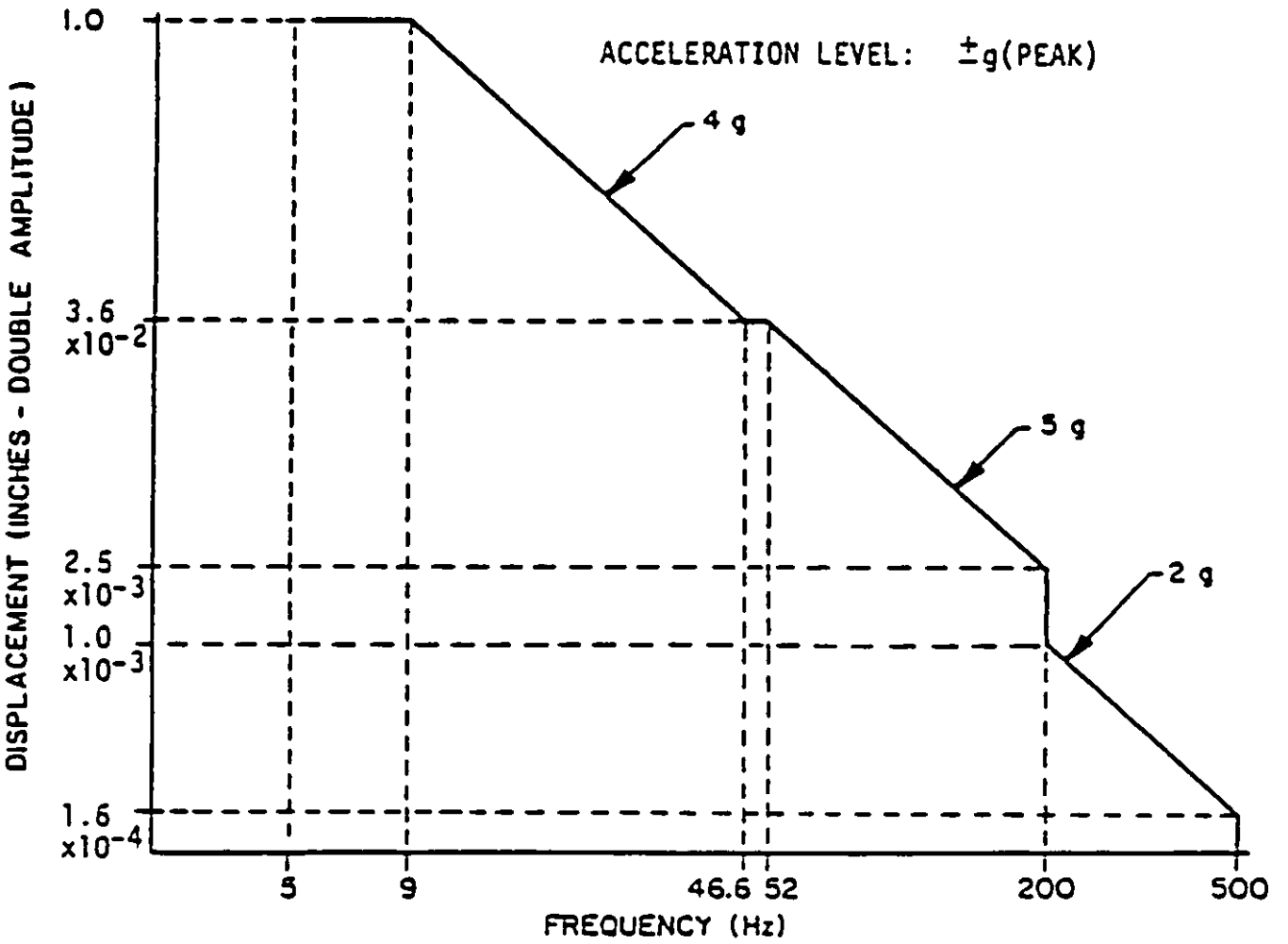


FIGURE 2. VIBRATION TEST PROFILE (X AXIS).

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